DawaaLink

Software Requirements Specification

Pharmacy Inventory Exchange Platform — v1.0

Document ID	DAWAA-SRS-001
Version	1.0
Standard	IEEE 830-1998 / ISO/IEC 29148:2018
Status	**Draft**
Date	April 2026
Tech Lead	Mazen (Callisto) Elnahal
Backend	Java / Spring Boot
Frontend	React (SPA) **[CONFIRMED]**
Database	PostgreSQL **[CONFIRMED]**
Geographic Scope	Greater Cairo
Target Delivery	June 2026
Institution	School of Computing and Data Science, Badya University
Supervisors	Prof Dr Mohamed Khafagy | Prof Dr Amira Mohamed | Prof Dr Amal Fouad | Prof Dr Marwa El Anany
Traces To	DAWAA-BRD-001 v1.0

Partially Resolved: Frontend (React), database (PostgreSQL), and API style (REST + WebSocket) are confirmed. Only the PDF generation library and drug reference dataset remain open. All other items are confirmed. Items marked **[TBD]** still require a team decision before development sprint 1 begins.

Contents

Introduction

Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for DawaaLink v1 — a web-based pharmacy inventory exchange platform. It is written in compliance with IEEE 830-1998 and ISO/IEC 29148:2018 and serves as the primary reference for the development team, test engineers, and academic supervisors throughout the development lifecycle.

This document traces directly to **DAWAA-BRD-001 v1.0** (Business Requirements Document). Every functional requirement herein maps to at least one BRD business requirement and one business objective.

Scope

DawaaLink enables independent pharmacies in **Greater Cairo** to exchange near-expiry dead stock through an automated, cash-free, multi-party barter system. The system automatically matches pharmacies holding excess stock of a medicine with pharmacies that need it, facilitates the confirmation workflow, and generates a formal Swap Record upon completion.

In scope for v1: Pharmacy registration, dead stock listing, automated matching engine (2-party and 3-party cycles), Swap Ticket generation, accept/decline workflow, 72-hour confirmation timer, PDF Swap Record generation, near-expiry alerting, role-based access control, audit logging, and KPI dashboard.

Out of scope for v1: Mobile native apps, platform-managed logistics, cash/payment processing, multi-branch chain management, EDA system integration, geographic scope outside Greater Cairo.

Definitions, Acronyms, and Abbreviations

Term	Definition
Swap Cycle	A closed barter loop: 2-party ($A \leftrightarrow B$) or 3-party ($A \rightarrow B \rightarrow C \rightarrow A$)
Swap Ticket	The offer card shown to both parties before any swap commitment is made
Swap Record	Auto-generated PDF documenting a completed swap for both parties' records
Dead Stock	Inventory at risk of expiring before sale
GTIN	Global Trade Item Number — universal pharmaceutical barcode identifier
EDA	Egyptian Drug Authority — national pharmaceutical regulator
OTC	Over The Counter — medication dispensed without prescription
RBAC	Role-Based Access Control
SKU	Stock Keeping Unit — unique product identifier in a pharmacy's inventory
REST	Representational State Transfer — API architectural style
JWT	JSON Web Token — stateless authentication mechanism

Term	Definition
DFD	Data Flow Diagram
SRS	Software Requirements Specification (this document)
BRD	Business Requirements Document (DAWAA-BRD-001)
TBD	To Be Decided — requires team decision before sprint 1

References

- DAWAA-BRD-001 v1.0 — DawaaLink Business Requirements Document
- DAWAA-BA-001 — DawaaLink Business Analysis Report
- DAWAA-SURVEY-001 — DawaaLink Consolidated Survey Responses (14 pharmacies)
- IEEE 830-1998 — Recommended Practice for Software Requirements Specifications
- ISO/IEC 29148:2018 — Systems and Software Engineering: Requirements Engineering
- BABOK Guide v3 — Business Analysis Body of Knowledge (IIBA)

Overview

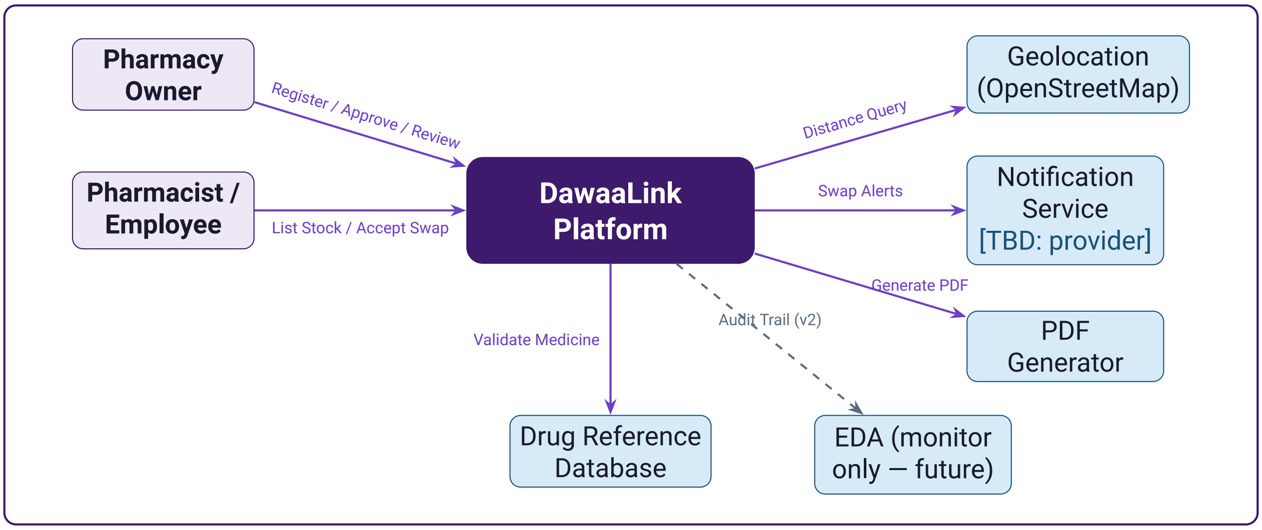
Section 2 provides the overall system description and context diagrams. Section 3 contains all functional requirements organised by subsystem. Section 4 covers non-functional requirements. Section 5 defines system and external interfaces. Section 6 presents all supporting UML diagrams. Section 7 is the appendix containing the ERD and data dictionary.

Overall Description

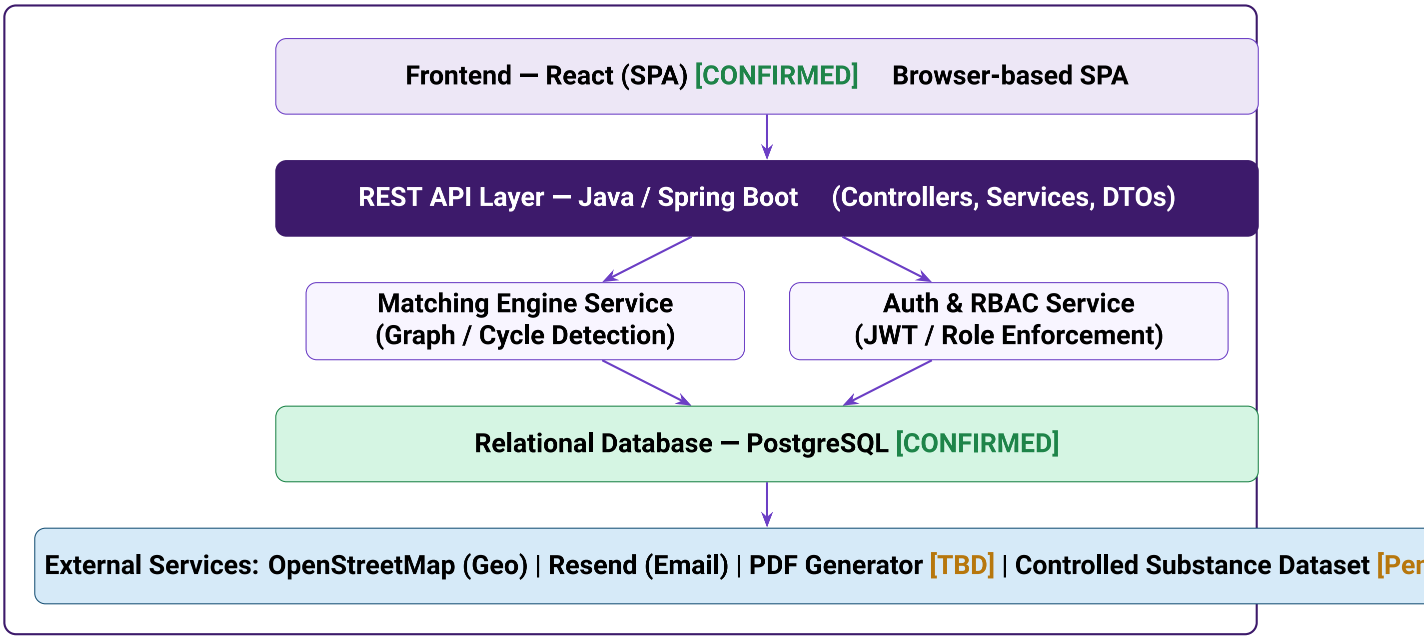
System Context

DawaaLink operates as a web application with a three-tier architecture: a browser-based frontend, a Java/Spring Boot REST API backend, and a relational database. It interacts with three external services: a geolocation API for distance calculation, a notification service for swap alerts, and a PDF generation library for Swap Records.

System Context Diagram (Level 0)



System Architecture



User Classes and Characteristics

User Class	Technical Level	Primary Actions	Access Level
Owner	Low–Medium	Approve registration; review dashboard; manage users	Full platform access
Pharmacist	Medium	List dead stock; review Swap Tickets; accept/decline swaps	Listing + swap management
Employee	Low	Data entry for dead stock items	Listing only; no swap decisions
System Admin	High	Platform administration; flag review; audit log access	Full backend access

Assumptions and Dependencies

- Pharmacies have internet access on desktop computers during business hours
- At least 20 pharmacies onboarded in Greater Cairo before launch (network density)
- Backend technology is fixed as Java / Spring Boot
- Frontend is React, database is PostgreSQL, API style is REST + WebSocket — all confirmed. Remaining open decisions (notification delivery, geolocation, PDF library, drug DB) must be finalised before sprint 1
- A drug reference database or GTIN dataset is available for medicine name validation
- Geolocation API access is provisioned before the matching engine is implemented

Functional Requirements

Requirements are organised by subsystem. Each requirement is stated as a **shall** statement, assigned a unique ID, priority level, and traceability to the BRD.

Authentication & Registration Subsystem

SRS-AUTH-01 **Critical** *Traces to: FR-01, FR-02*

The system **shall** provide a pharmacy registration form collecting: pharmacy name, district/area (Greater Cairo), owner full name, commercial registration number, and primary pharmacist contact (name + phone). Submitted registrations **shall** be assigned status `PENDING` until the owner explicitly activates the account.

Input: Registration form fields
Processing: Validate all fields; check for duplicate commercial registration number; assign `PENDING` status; send activation prompt to owner
Output: Account created with status `PENDING`; owner receives activation notification
Error: Duplicate registration number ▯ reject with message; missing field ▯ inline validation error

SRS-AUTH-02 **Critical** *Traces to: FR-02*

The system **shall** implement role-based access control (RBAC) with three user roles: **Owner**, **Pharmacist**, and **Employee**. Roles **shall** be enforced on every API endpoint. Authentication **shall** use JWT tokens with a configurable expiry (default: 8 hours).

Role permissions:

- **Owner:** All actions including user management, full swap history, KPI dashboard
- **Pharmacist:** List dead stock, view matches, accept/decline swaps, view own swap history
- **Employee:** List dead stock items only; cannot view matches or make swap decisions

Error: Unauthorised role attempting restricted action ▯ HTTP 403 Forbidden

SRS-AUTH-03 **Critical** *Traces to: FR-01*

The system **shall** prevent any swap-related action (listing, matching, accepting) from a pharmacy account with status `PENDING` or `SUSPENDED`.

Error: Attempt by `PENDING` account ▯ HTTP 403 with message: "Account activation required. Please ask your pharmacy owner to activate this account."

Dead Stock Listing Subsystem

SRS-LIST-01 **Critical** *Traces to: FR-03*

The system **shall** provide a dead stock listing form with the following mandatory fields:

- **Medicine Name** (text; validated against drug reference database)
- **Batch Number** (alphanumeric; free text)
- **Expiry Date** (date picker; must be a future date)
- **Quantity Available** (positive integer)
- **Storage Condition** (enum: `ROOM_TEMP` / `COLD_CHAIN`)

- **Unit** (enum: `BOX` / `STRIP` / `VIAL` / `PIECE`)

Optional fields: unit price (EGP), notes.

Error: Missing mandatory field □ inline validation; past expiry date □ reject with message

SRS-LIST-02 **Critical** *Traces to: FR-04*

The system **shall** validate medicine names against a drug reference database upon submission. If the medicine name matches a controlled substance category (schedule drugs, narcotics), the listing **shall** be assigned status `FLAGGED` and **shall not** enter the matching pool. A system administrator **shall** be notified of the flagged listing.

If the medicine name matches `COLD_CHAIN` storage requirement and the user selects `ROOM_TEMP`, the system **shall** display a warning and require explicit confirmation.

Error: Controlled substance detected □ status `FLAGGED`; submitting pharmacy notified

SRS-LIST-03 **High** *Traces to: FR-05*

The system **shall** calculate days-to-expiry for every active listing on a daily scheduled job. Items where days-to-expiry falls within a configurable threshold (default: 90 days) **shall** be flagged with a colour-coded urgency badge:
RED ($<$30 days) | **AMBER** (30–60 days) | **YELLOW** (60–90 days)

Flagged items **shall** appear in a dedicated Near-Expiry panel on the pharmacy dashboard.

Matching Engine Subsystem

SRS-MATCH-01 **Critical** *Traces to: FR-06*

The system **shall** run a matching algorithm (triggered on new listing creation and on a periodic scheduled job, minimum every 30 minutes) that identifies pharmacies within a configurable geographic radius (default: 10 km, maximum: 25 km within Greater Cairo) where:

- Pharmacy A lists medicine X with status `ACTIVE`
- Pharmacy B has medicine X on its wishlist or has previously listed a need for it

The algorithm **shall** detect both 2-party cycles ($A \leftrightarrow B$) and 3-party cycles ($A \to B \to C \to A$). Items with status `FLAGGED`, `LOCKED`, `COMPLETED`, or `CANCELLED` **shall** be excluded.

Algorithm: Directed graph where nodes = pharmacies, edges = supply/demand overlap. Cycle detection via Depth First Search (Johnson's Algorithm for 3-party cycles).

SRS-MATCH-02 **Critical** *Traces to: FR-07*

Upon cycle detection, the system **shall** atomically lock all involved inventory items (status: `LOCKED`) across all parties simultaneously before issuing any Swap Tickets. The lock operation **shall** be implemented as a database transaction to prevent concurrent double-matching.

Error: Lock acquisition fails (race condition) □ retry once; if still fails □ skip this cycle and re-queue for next matching run

SRS-MATCH-03 **Critical** *Traces to: FR-07*

If any party declines a swap or the 72-hour timer expires, the system **shall** atomically release all `LOCKED` items back to `ACTIVE` status for all parties in the cycle simultaneously.

Swap Ticket & Confirmation Subsystem

SRS-SWAP-01 **Critical** *Traces to: FR-08*

Upon successful lock, the system **shall** generate and deliver a Swap Ticket to each party in the cycle. The Swap Ticket **shall** display:

- Medicine name, batch number, expiry date, quantity offered, storage condition
- Offering pharmacy name and district
- Distance between pharmacies (km)
- Days remaining until expiry (colour-coded)
- Swap Ticket expiry countdown (72 hours from issuance)
- What the accepting pharmacy will receive in return

Delivery: In-app dashboard notification via WebSocket (`ws://.../ws/notifications`). **[CONFIRMED]**

SRS-SWAP-02 **Critical** *Traces to: FR-09*

The system **shall** present Accept and Decline actions on the Swap Ticket accessible to Owner and Pharmacist roles only. A swap cycle **shall** advance to `CONFIRMED` status only when **all** parties have accepted. Any single Decline **shall** immediately trigger `CANCELLED` for the entire cycle and release all locked items.

Error: Employee role attempts to accept/decline → action blocked; HTTP 403

SRS-SWAP-03 **High** *Traces to: FR-10*

The system **shall** enforce a 72-hour confirmation timer starting from Swap Ticket issuance. A reminder notification **shall** be sent to non-responding parties at 24 hours remaining. At timer expiry with no response from any party, the system **shall** automatically cancel the cycle and release all locked items.

Implementation: Scheduled job polling pending swaps every 15 minutes.

SRS-SWAP-04 **Critical** *Traces to: FR-09*

Upon `CONFIRMED` status, the system **shall** generate transfer instructions for each party specifying: what to send, to whom, and the counterparty's pharmacy name and district. The system **shall** then await receipt confirmation from both parties before marking the swap `COMPLETED`.

States after confirmation: `CONFIRMED` → `IN_TRANSFER` → `COMPLETED`

Swap Record Subsystem

SRS-REC-01 **Critical** *Traces to: FR-11*

Upon swap `COMPLETED` status, the system **shall** automatically generate a PDF Swap Record within 60 seconds containing:

- Unique Swap ID (format: `DAWAA-[YYYYMMDD]-[6-digit-seq]`)

- Date and time of completion (CLT timezone)
- For each leg: sending pharmacy name & district, receiving pharmacy name & district, medicine name, batch number, expiry date, quantity transferred, direction (Sent/Received)

The record **shall** be accessible from the Swap History dashboard and downloadable at any time. Filename format: `DAWAA-[SwapID]-[Date].pdf`

SRS-REC-02 **High** *Traces to: FR-12*

The system **shall** maintain an immutable audit log recording every state transition for every entity (inventory items, swap cycles, user accounts). Each log entry **shall** contain: entity type, entity ID, previous state, new state, actor user ID, timestamp, and IP address. Log entries **shall** be write-only — no delete or update operations permitted on the audit log table.

Dashboard & Reporting Subsystem

SRS-DASH-01 **High** *Traces to: FR-13*

The system **shall** present a pharmacy dashboard as the default post-login view displaying:

- **KPI Cards:** Total items listed, matches found this week, total swaps completed, estimated value recovered (EGP)
- **Near-Expiry Panel:** Items flagged within threshold, sorted by urgency
- **Active Swap Matches:** Pending Swap Tickets awaiting response
- **Recent Swap History:** Last 10 swaps with status, direction, and counterparty

KPI values **shall** reflect real-time data. Dashboard **shall** be accessible to Owner and Pharmacist roles.

Non-Functional Requirements

Performance

- **SRS-NFR-01:** API response time for all non-matching endpoints **shall** be $\leq$500 ms at the 95th percentile under normal load.
- **SRS-NFR-02:** The matching engine **shall** complete a full matching run for up to 500 active pharmacies in $\leq$30 seconds.
- **SRS-NFR-03:** PDF Swap Record generation **shall** complete within 60 seconds of swap completion.

Security

- **SRS-NFR-04:** All data **shall** be encrypted in transit using TLS 1.3 minimum.
- **SRS-NFR-05:** All sensitive data at rest **shall** be encrypted using AES-256.
- **SRS-NFR-06:** All API endpoints **shall** validate JWT tokens and enforce RBAC on every request.
- **SRS-NFR-07:** The system **shall** implement rate limiting on authentication endpoints (max 10 attempts/minute per IP).
- **SRS-NFR-08:** No pharmacy's inventory data **shall** be accessible to any user outside of matched swap partners and system administrators.

Availability & Reliability

- **SRS-NFR-09:** The platform **shall** maintain $\geq$99% uptime during business hours (08:00–22:00 CLT, 7 days a week).
- **SRS-NFR-10:** The matching engine scheduled job **shall** be idempotent — running it twice **shall** not create duplicate swap cycles.
- **SRS-NFR-11:** The system **shall** handle concurrent matching requests without data corruption via database-level locking.

Usability

- **SRS-NFR-12:** Core workflows (list item, view match, accept swap) **shall** be completable in $\leq$3 minutes by a first-time user without training.
- **SRS-NFR-13:** The platform **shall** be fully functional on Chrome, Firefox, and Edge (latest 2 versions) on Windows desktop.
- **SRS-NFR-14:** The platform **shall** be mobile-responsive with no broken layouts on screens $\geq$375px wide.
- **SRS-NFR-15:** All UI text, error messages, labels, and notifications **shall** be available in both **English** and **Arabic** (RTL supported).

Scalability

- **SRS-NFR-16:** The system architecture **shall** support $\geq$500 concurrent pharmacy accounts without performance degradation below SRS-NFR-01 thresholds.
- **SRS-NFR-17:** The matching engine **shall** be implemented as a separate service component that can be scaled independently from the REST API layer.

Maintainability

- **SRS-NFR-18:** All REST API endpoints **shall** be documented using OpenAPI 3.0 (Swagger).
- **SRS-NFR-19:** The codebase **shall** maintain $\geq$70% unit test coverage on service layer classes.
- **SRS-NFR-20:** Database schema changes **shall** be managed via migration scripts (e.g. Flyway or Liquibase).

External Interface Requirements

User Interface

- The system **shall** be delivered as a Single Page Application (SPA) with a persistent left sidebar navigation, top header bar, and main content area.
- Sidebar navigation items: Dashboard, My Dead Stock, Swap Matches, Active Swaps, Swap History, Pharmacy Profile, Settings.
- Primary colour: `#6C3FC5` (purple). Background: white. Accent: `#EDE7F6`.
- Typography: Roboto or Inter. No serif fonts.

API Interface

- The backend **shall** expose a RESTful API over HTTPS.
- All request and response bodies **shall** use JSON (Content-Type: `application/json`).
- API versioning **shall** be implemented via URL prefix: `/api/v1/...`
- Authentication **shall** use Bearer JWT tokens in the `Authorization` header.
- The system **shall** implement WebSocket support (`/ws/notifications`) for real-time swap notifications. **[CONFIRMED]**

External Service Interfaces

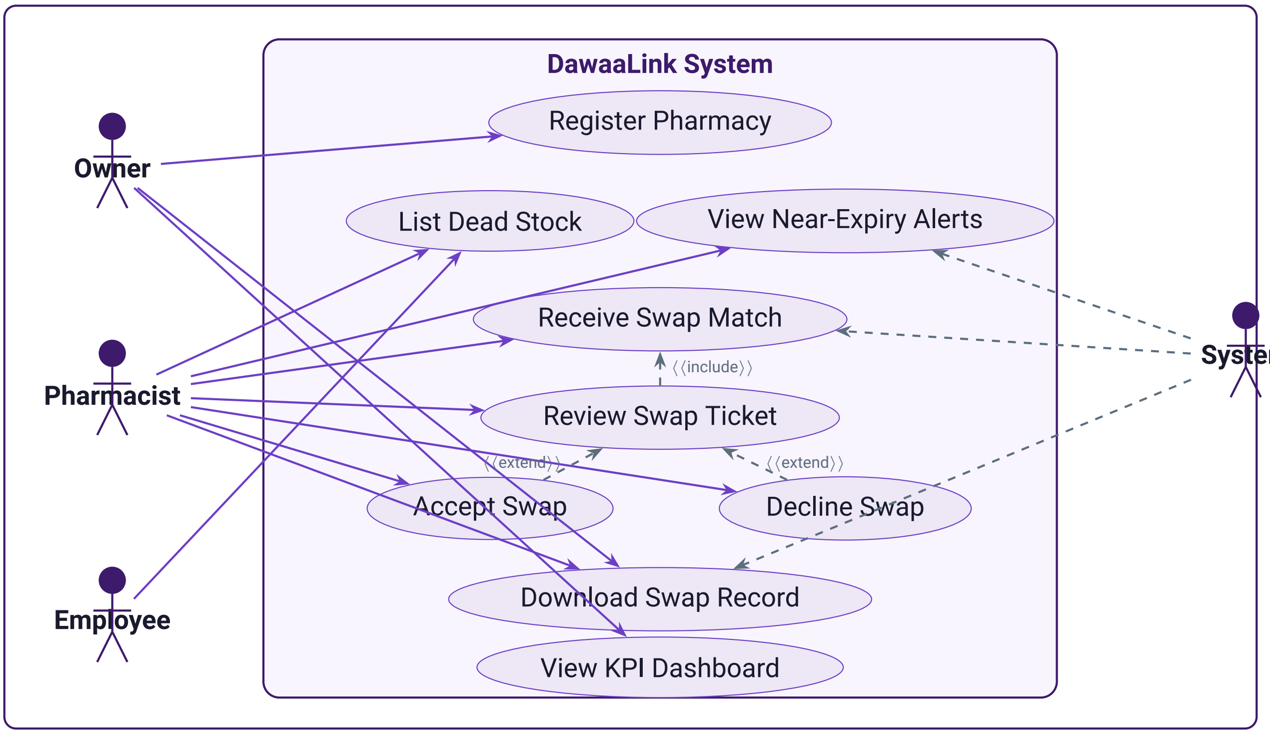
Service	Provider	Used For	Decision Status
Geolocation API	OpenStreetMap / Nominatim (free, open-source)	Distance calculation between pharmacies for matching radius	**[CONFIRMED]**
Notification Service	Email SMTP or SMS Gateway	Swap Ticket delivery; 24h reminder; completion notification	**[TBD]**
PDF Generator	iText, Apache PDFBox, or JasperReports	Swap Record PDF generation	**Team to confirm**
Drug Reference DB	Team-provided controlled substance dataset	Controlled substance flagging (FR-04); general medicine name validation stubbed for v1	**Dataset pending — provided by Tech Lead**

Hardware Interface

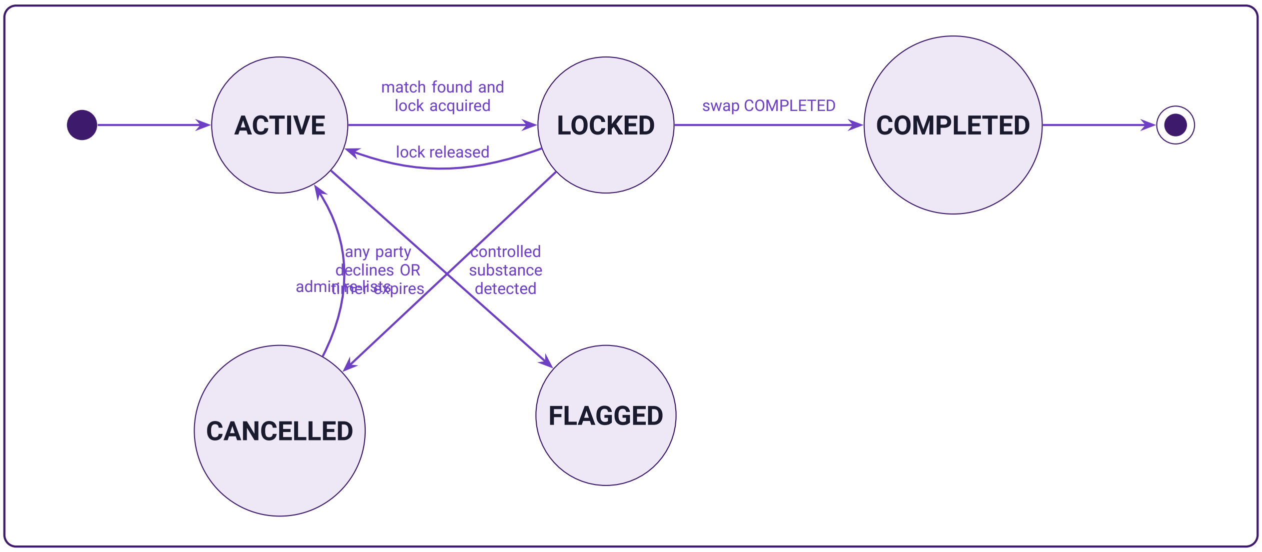
The system requires no dedicated hardware. It **shall** run on any cloud-hosted server environment supporting Java 17+ and a relational database. Desktop PC with internet access is the minimum client requirement.

System Diagrams

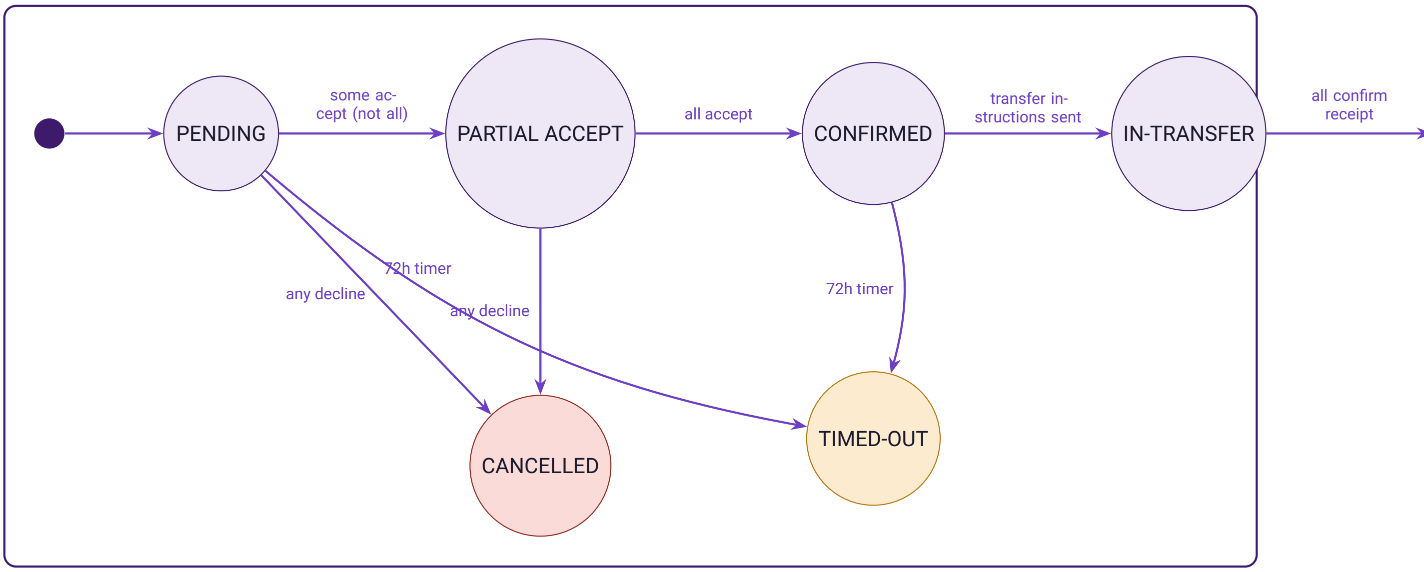
Use Case Diagram



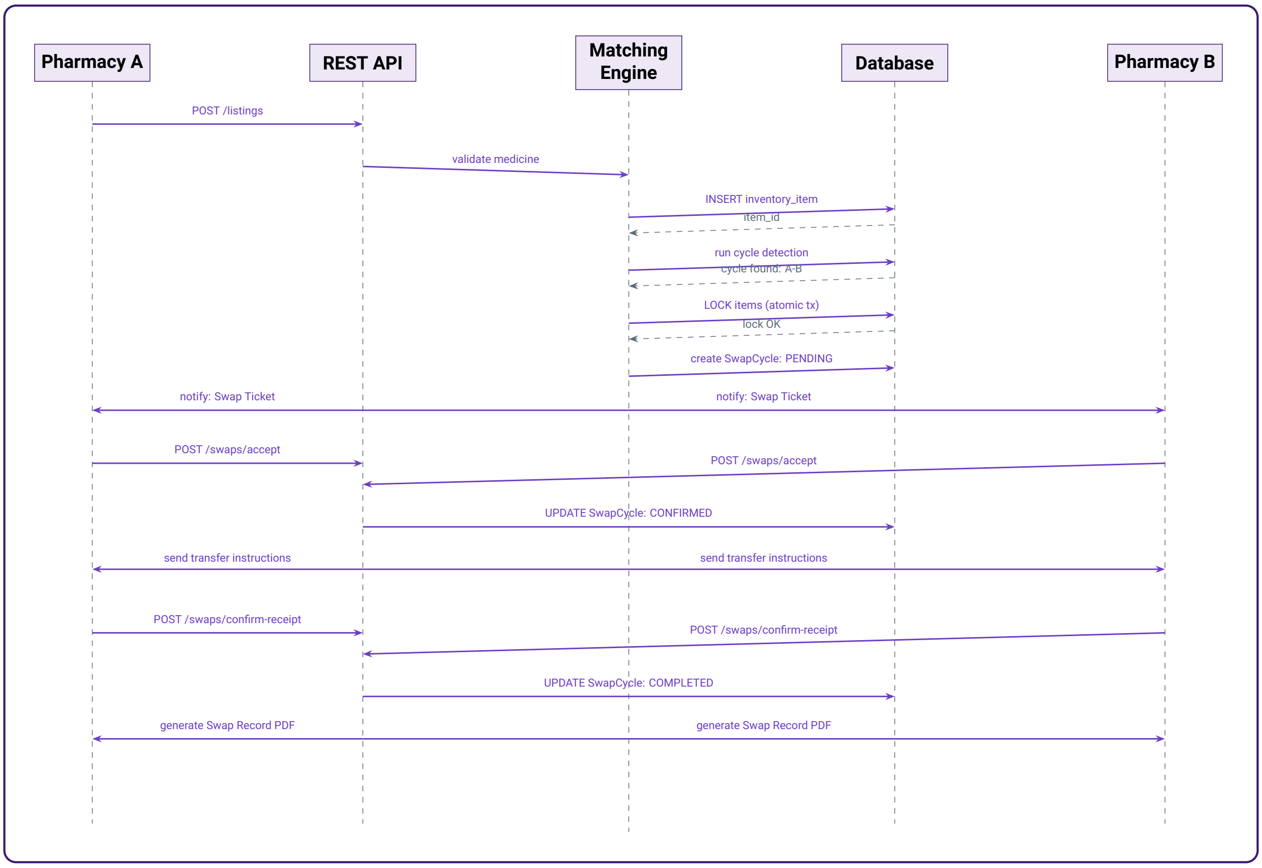
Inventory Item State Machine



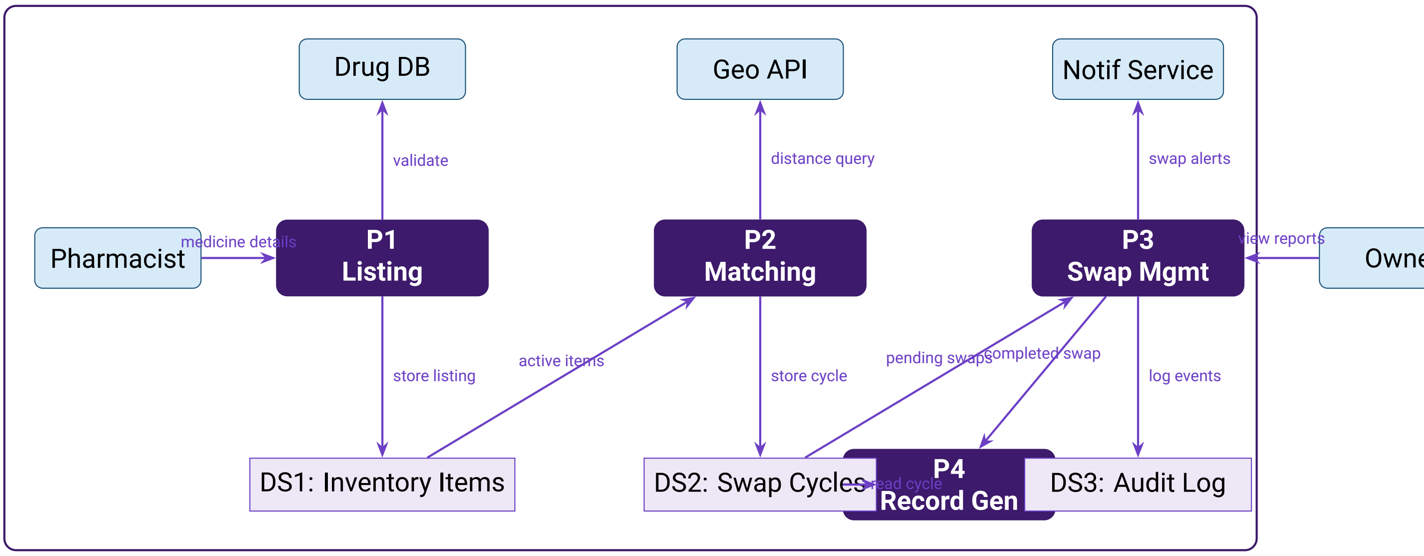
Swap Cycle State Machine



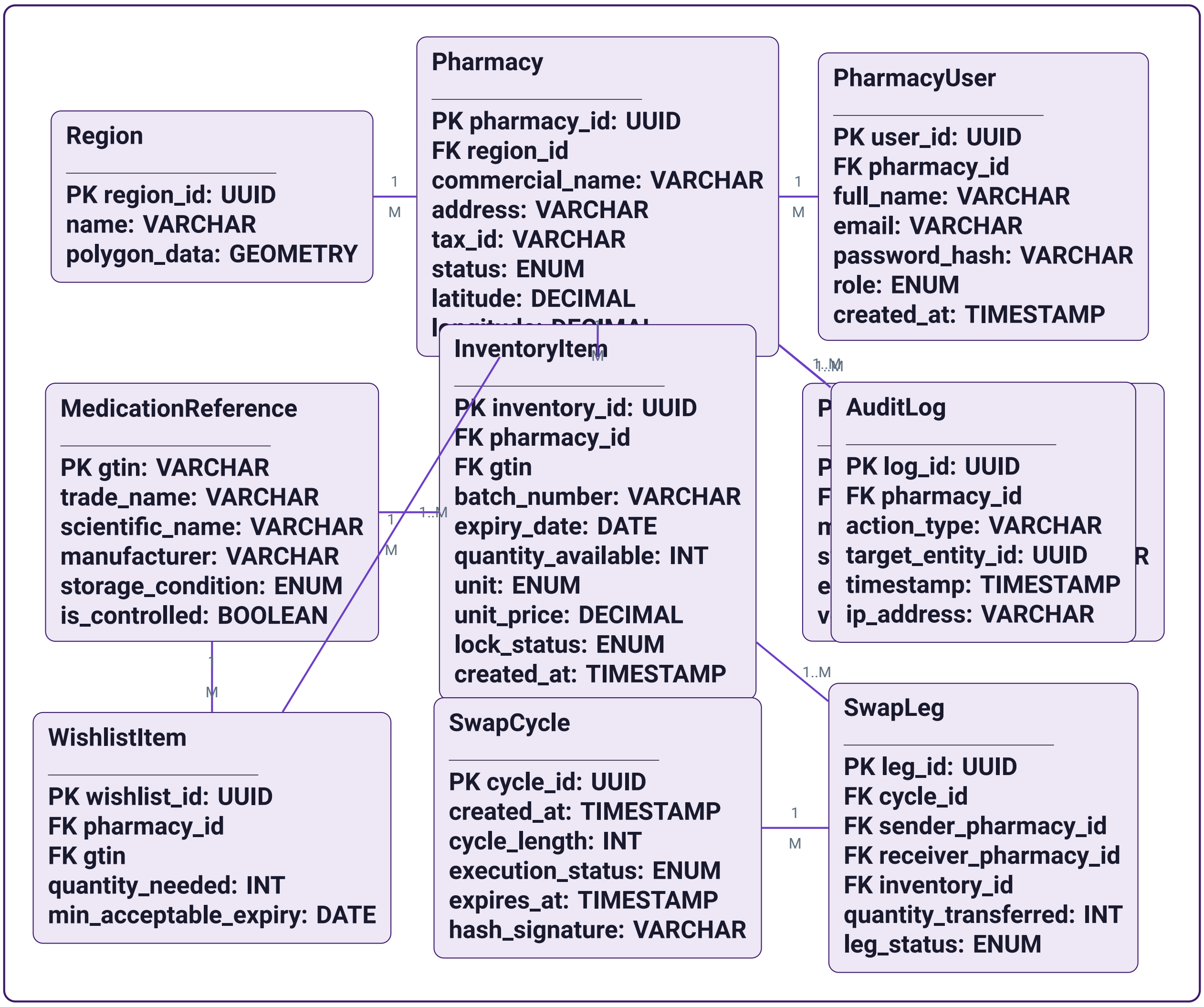
Sequence Diagram — Matching & Swap Confirmation Flow



Data Flow Diagram — Level 1


Drug DB
Geo API
Notif Service
validate
distance query
swap alerts
Pharmacist
medicine details
P1
Listing
P2
Matching
P3
Swap Mgmt
view reports
Owne
store listing
active items
store cycle
pending swaps
completed swap
log events
DS1: Inventory Items
DS2: Swap Cycles
read cycle
P4
Record Gen
DS3: Audit Log

Appendix A — Entity-Relationship Diagram



Appendix B — Requirements Traceability Matrix

SRS ID	Requirement	BRD FR	BRD BR	Priority	Test Case
SRS-AUTH-01	Pharmacy registration with owner approval	FR-01, FR-07	BR-07	Critical	TC-REG-01
SRS-AUTH-02	Role-based access control (RBAC) + JWT	FR-02	BR-07	Critical	TC-AUTH-01
SRS-AUTH-03	Block PENDING / SUSPENDED accounts	FR-01	BR-07	Critical	TC-AUTH-02
SRS-LIST-01	Dead stock listing form with mandatory fields	FR-03	BR-01	Critical	TC-LIST-01
SRS-LIST-02	Controlled substance detection + flagging	FR-04	BR-05	Critical	TC-CTRL-01
SRS-LIST-03	Near-expiry alerting with colour badges	FR-05	BR-01	High	TC-ALERT-01
SRS-MATCH-01	Matching engine: 2-party and 3-party cycles	FR-06	BR-02	Critical	TC-MATCH-01
SRS-MATCH-02	Atomic inventory lock on match	FR-07	BR-02	Critical	TC-LOCK-01
SRS-MATCH-03	Atomic lock release on decline / timeout	FR-07	BR-02	Critical	TC-LOCK-02
SRS-SWAP-01	Swap Ticket generation with all required fields	FR-08	BR-03	Critical	TC-TICKET-01
SRS-SWAP-02	Accept / Decline workflow; all-party confirmation	FR-09	BR-03	Critical	TC-CONF-01
SRS-SWAP-03	72-hour timer with 24h reminder and auto-cancel	FR-10	BR-02	High	TC-TIMER-01
SRS-SWAP-04	Transfer instructions + receipt confirmation	FR-09	BR-03	Critical	TC-TRANSFER-01
SRS-REC-01	Auto-generated PDF Swap Record	FR-11	BR-04	Critical	TC-REC-01
SRS-REC-02	Immutable audit log	FR-12	BR-04, BR-06	High	TC-AUDIT-01
SRS-DASH-01	KPI dashboard + near-expiry panel + swap history	FR-13	BR-01, BR-04	High	TC-DASH-01

Confirmed Decisions:
React (Frontend) ✓ PostgreSQL (Database) ✓ REST + WebSocket (API) ✓

Still Open Before Sprint 1:

1. PDF generation library (iText / PDFBox / JasperReports)
2. Drug reference database — controlled substance dataset to be provided by Tech Lead before sprint 2

Each decision should be recorded as an addendum to this document and reflected in the architecture diagram.

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